

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1. Marginal PMF

Definition

The key idea:

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

Notation

If joint PMF is $p_{X,Y}(x, y)$, then:

Marginal PMF of X :

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

Marginal PMF of Y :

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Marginal PMF from Joint PMF Table

Step-by-step procedure:

Start with the joint PMF table

To find $p_X(x)$:

- Fix x
- Sum across all y

To find $p_Y(y)$:

- Fix y
- Sum across all x

Example (as per slides)

Given: Joint PMF

Solution Steps:

Apply definition:

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

Similarly:

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Compute values by summing corresponding rows/columns

Conclusion:

Marginal PMFs are obtained directly by summation over the joint PMF.

2. Correlation

Definition

A basic way to measure joint behavior:

$$R_{XY} = E[XY]$$

Interpretation

It gives the average value of product XY

Called a raw correlation measure

Important Observations

Depends on scale of X and Y

Computed about the origin (not mean)

Mixes:

- Mean values
- Spread
- Dependence

Therefore:

Not a clean measure of dependence

Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

Transition to Covariance

To remove mean effects:

We center variables $\rightarrow$ leads to covariance

3. Covariance

Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Step-by-Step Interpretation

Subtract mean from each variable

Measure how deviations vary together

Meaning:

Positive $\rightarrow$ move in same direction

Negative $\rightarrow$ move in opposite direction

4. Covariance: Algebraic Form

Start with:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Expand:

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

Apply linearity of expectation:

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

Simplify:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

5. Properties of Covariance

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

3. Linearity with Constants

(Covariance behaves linearly when constants are involved)

4. Independence Property

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

Important:

Zero covariance $\Rightarrow$ uncorrelated

But NOT necessarily independent

6. Covariance Example 1

Method (as given)

Use formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Substitute:

$$E[XY]$$

$$E[X]$$

$$E[Y]$$

Simplify step-by-step

Final result obtained after substitution

7. Pearson's Correlation Coefficient

Definition

Measures effect of change in one variable when the other changes

Key Points

Measures strength and direction of linear relationship

Normalized version of covariance

Unit-free

8. Covariance Example 2 (Continuous Case)

Given:

Joint PDF $f_{X,Y}(x, y)$

Steps:

Compute:

$$E[X], E[Y], E[XY]$$

Use:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Perform integration over valid region

Substitute results

Final covariance obtained

9. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is jointly Gaussian if:

Joint density has bivariate Gaussian form

Marginals are also Gaussian

Interpretation

Models continuous random variables influenced by many small effects

Real-Life Applications

Signal processing

Communication systems

Noisy measurements